



Generative Agent UAT Overview

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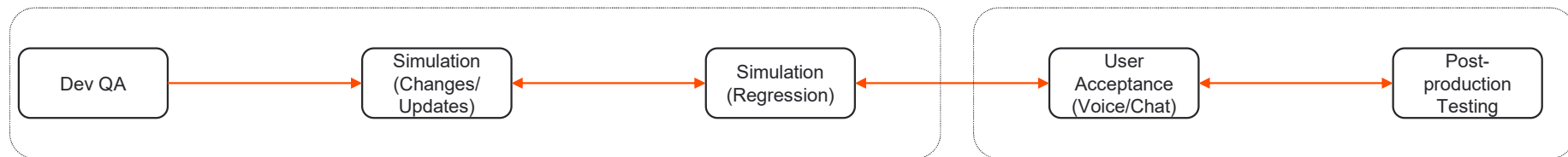
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Generative Agent: Testing Overview

The goal in testing generative agents is to ensure they do not harm to the user we check for:

- Accuracy: Ensuring all information provided is correct and up to date.
- Consistency: Maintaining a consistent tone and approach in all interactions.
- Compliance: Adhering to all relevant regulations and company policies.



As GA Engineers and Conversational Designers make updates and changes, they simulate conversations to validate their work.

All changes are grouped for a Release and Simulations are created to validate changes, updates, and connections as applicable.

A set of business specified cases are simulated where both the interaction and the expected response are provided. As simulations run, the GA responds and a different LLM reviews the output and groups them into Pass or Fail. Changes are made to the GA until all simulations pass.

Business users complete a specific number of test cases through either or both channels. The results are graded with Never, Must, or Should. Must and Never issues are resolved and retested prior to release.

Once the changes are moved to production, all interactions are monitored and reviewed for a pre-determined time to ensure no impacts prior to BAU QA and review.

Generative Agent: UAT Test Cases



Test Case Creation

When creating test cases, it is important that scenarios are tied directly to the stated business requirements, like how we evaluate human agents. If the agent has not been provided the necessary rules, information, or limits, it cannot be expected to perform correctly and may fail entirely in situations outside its defined scope. Some edge cases can be addressed during testing as long as they remain within the business requirement boundaries, while others may need to be identified and refined over time in production, just as we would with a new human agent.

Test Case Goals

1. Adherence to Business Requirements: Follow predefined policies, procedures, and assigned limits to ensure consistency and adherence to company business rules.
2. Factual Data Output: Responses must be rooted in factual, accurate data. While conversational elements may vary, factual data should remain consistent and reliable.
3. Evaluating Conversational Quality: Ensure that conversations are coherent, contextually appropriate, and provide valuable information to the customer.

Generative Agent: Test Case Grading Criteria



Must / Never / Should ratings help evaluate Generative AI responses based on business-critical behavior, not just exact wording. Since responses can vary conversationally, this approach focuses testing on whether the agent consistently performs required actions, identifies unacceptable behavior, and considers conversational best practice.

Never: These are actions an agent (live or generative) should never take or do, such as sharing confidential company information, violating privacy or HIPPA laws, providing financial, medical, or legal advice, or making discriminatory or offensive statements. Cases graded as Never would be considered a defect and require correction and retesting before deployment.

Must: These are non-negotiable aspects that the agent (live or generative) must always perform, such as confirming customer identity before sharing sensitive information, providing accurate product details, and sharing accurate information, including quoting data elements such as dollar figures, dates, or required disclosures. Any cases graded as Must would be considered a defect and require correction and retesting before deployment.

Should: These are best practices that the agent (live or generative) should generally follow to enhance the customer experience. Examples include using polite language, providing additional helpful resources, and maintaining a conversational speed and tone that aligns with the brand voice. Any cases graded with Should are effectively quality optimization or tuning opportunities. The Agent could move to production and would require continual iteration and tuning.

Pass: The interaction meets production standards. The agent completes the intended task accurately, follows all required behaviors, avoids prohibited behaviors, and delivers an acceptable customer experience.

Generative Agent: UAT Sample Methodology



Sigma-based sampling methodology uses statistical confidence modeling to determine an appropriate number of test cases needed to evaluate Generative Agent performance with a reliable level of accuracy and consistency. The approach applies standard quality and variance assumptions to balance testing precision against execution effort. Rather than testing every possible interaction, a representative sample is selected that provides confidence the results reflect overall production behavior.

The methodology helps:

- Validate performance consistently across interaction types
- Reduce testing effort while maintaining statistical reliability
- Support objective quality measurement and reporting
- Identify trends, defects, and edge cases prior to production
- Align AI testing practices with established quality assurance standards used for human agent evaluations

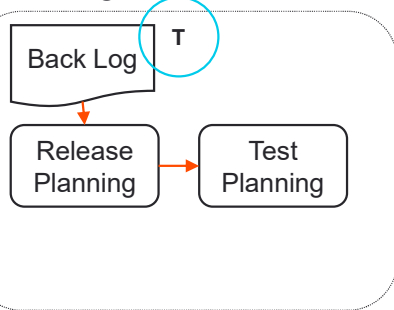
Example Sample Size:

Human to GA					7 days* 7 hours				
Population	Z-Score	Variability	Margin	Sample	Test Case Length (M)	Est Test Hours	Avail Test Hours	# Testers	Days to Test
445	1.96	0.50	5%	206	5	17.20	49	0.35	2.46
445	1.96	0.50	7%	136	5	11.36	49	0.23	1.62

Pre and Post Release - Gen Agent Monitoring

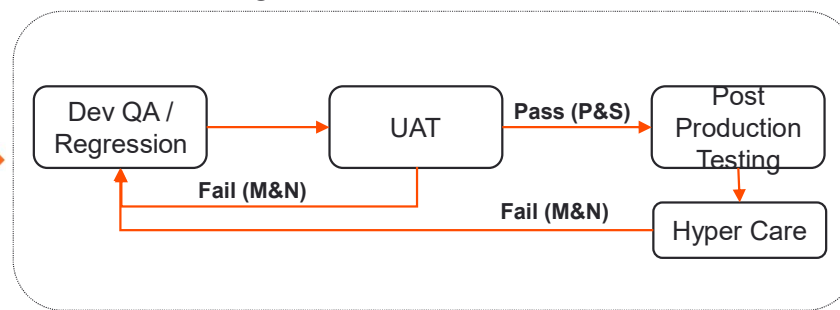
As updates and changes made to the Gen Agent handoffs between teams are important. Understanding when and what teams are responsible for reduces effort and ensures ongoing monitoring.

Tuning & Increments



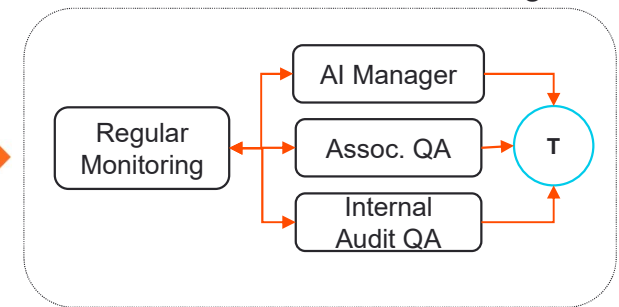
- Release Planning is a combined review by all relevant working parties to ensure value is added to the GenAgent.
- Test Planning is specific to the planned release as the complexity of the release will determine the complexity of the testing required.

Testing Framework



- Conversational Designer and or GA Engineer will test their updates. They will also complete prescribed Regression/Simulation testing prior to UAT.
- UAT testing is a combination of Simulation and Live agent interaction through voice or chat and is conducted by the business team(s) or delegates leveraging the Must/Never/Should framework.
- Post Production Testing duration and volume is determined in release planning based on the complexity of the release. Any Issues or Opportunities would be resolved through hyper care.
- Successful exit of Post- production testing would signal the start of BAU reviews by the relevant business teams.

Business As Usual Monitoring



- FEP Manager performs GenAgent spot QA, runs manual simulations to validate scenarios, and combines opportunities from Audit and Assoc. for review with Product and Conversational Designer.
- Assoc. QA samples and audits based on current processing rules.
- Internal Audit / QA – Samples and provides feedback based on interaction volume.